agronomy

MDPI

Article

Untargeted Metabolite Profiling of *Camellia tetracocca*'s Response to an *Empoasca onukii* Attack Using GC-MS and LC-MS

Ni Zhang [1,2], Weiwen Tan [1,2], Guimei Luo [1,2], Tianyi Pu [1,2], Jinqiu Wang [1,2], Xianhu Zhang [3] and Yuehua Song [1,2,*]

[1] School of Karst Science, Guizhou Normal University, Guiyang 550001, China
[2] State Engineering Technology Institute for Karst Desertification Control, Guiyang 550001, China
[3] State Pubai Forest Farm of Pu'an County Forestry Bureau, Xingyi 562400, China
* Correspondence: songyuehua@163.com

Abstract: *Camellia tetracocca*, a kind of tea with local popularity, is unique to southwest China, where it has an important natural heritage and cultural heritage. However, the tea plant and its sprout are frequently attacked on a large-scale by *Empoasca onukii*. The metabolic mechanisms of the unique plant for defending against these pest insects are unclear. Therefore, we used untargeted gas chromatography–mass spectrometry (GC-MS) and high performance liquid chromatography–mass spectrometry (LC-MS) to compare the metabolite profiles between *E. onukii*-attacked leaves and healthy leaves. Using GC-MS, 56 metabolites (24 up-regulated, 32 down-regulated) were preliminarily identified. Additionally, 576 metabolites (280 up-regulated, 287 down-regulated) were rudimentarily identified with LC-MS. Differentially abundant metabolites were mainly enriched in the biosynthesis of specialized metabolites. Fourteen accumulated specialized metabolites are related to insect resistance. Mainly, momordicin I and arabidopside B are reportedly involved in the resistance to the insect. Therefore, we conjectured that the accumulation of momordicin I and arabidopside B is involved in the *C. tetracocca*'s resistance to *E. onukii*. Our results indicate that these specialized metabolites may be served as candidate biocontrol agents against the pest of *E. onukii* of *C. tetracocca* located in the State-owned Pubai Forest Farm.

Keywords: *Camellia tetracocca*; *Empoasca onukii*; metabolomics; GC-MS; LC-MS; Karst

check for updates

Citation: Zhang, N.; Tan, W.; Luo, G.; Pu, T.; Wang, J.; Zhang, X.; Song, Y. Untargeted Metabolite Profiling of *Camellia tetracocca*'s Response to an *Empoasca onukii* Attack Using GC-MS and LC-MS. *Agronomy* **2023**, *13*, 791. https://doi.org/10.3390/agronomy13030791

Academic Editor: Alessandra Carrubba

Received: 25 January 2023
Revised: 27 February 2023
Accepted: 7 March 2023
Published: 9 March 2023

Copyright: © 2023 by the authors. Licensee MDPI, Basel, Switzerland. This article is an open access article distributed under the terms and conditions of the Creative Commons Attribution (CC BY) license (https://creativecommons.org/licenses/by/4.0/).

1. Introduction

Camellia tetracocca Zhang, an ancient tea tree, belongs to the *Camellia* L., subgen. Thea, ser. Quiquelocularis, which is a unique tea variety discovered in Guizhou by Mr. Hongda Zhang in 1981 [1,2]. *C. tetracocca* species in Pu'an are mainly distributed in Majiaping Village, Pubai Forest Farm, other areas, and include the wild type, cultivated type, and transitional type [3,4]. *C. tetracocca* arrogate the characteristics of low output and high quality on accounting of its primitive growth environment and slow growth rate, which makes it the best raw material for processing and manufacturing high-grade black tea [5]. The unique geographical advantages of Pu'an have accelerated the development of the ancient tea tree. However, because of its narrow and special distribution, the tea plant is still only planted and consumed locally today [6]. Unfortunately, for a multitude of reasons, including excessive picking, environmental degradation, large-scale outbreaks, pest damage and plant disease have caused the frequent death of *C. tetracocca*.

The growth of *C. tetracocca* is usually affected by miscellaneous herbivorous insects. The tea green leafhopper is the most universal pest in the tea field with the longest life cycle, making it difficult to control [7]. The tea green leafhopper, *Empoasca onukii* (Hemiptera: Cicadellidae), is the most disruptive pest across tea plantations. The nymphs and adults pierce and suck the sap of tender tea shoots [8]. Moreover, adult females lay their eggs in

these shoots, thus leading to irreversible damages [9,10]. *E. onukii* is excellently adapted to diverse tea varieties, both physiologically and biochemically [11]. *C. tetracocca* in Pu'an offers a stable habitat and food source for *E. onukii*, but the leafhopper causes serious damage to *C. tetracocca*, including tea buds withered to exsiccation and the apex of the tea shoot displaying "Hopperburn" [12,13], thus a large reduction in tea production or even tea tree death occurs [10].

Plants are intelligent organisms that generate numerous morphological, physiological, biochemical, and molecular mechanisms to protect themselves from insect herbivores [14–16]. In addition to modifications at the transcriptome and proteome, secondary metabolite synthesis is also one of the most important defense mechanisms to maintain the growth of the plant itself under insect damage [17,18]. Secondary metabolites have no direct function in the elementary life process of plants [19,20], but they demonstrate significant value in habituation and shielding against phytophagous insects [21]. These metabolites are mainly divided into four categories: terpenoids, phenols, nitrogen-containing and sulfur-containing compounds [22–25]. In addition, plants can release volatile organic compounds (VOCs) to prohibit herbivores or allure natural enemies of pests when damaged by herbivorous insects [26]. Benzoic acid derivative levels were increased due to damage from thrips and lepidopterans in the foliage of the field-grown soybean cultivars [27]. Rice (*Oryza sativa*) plants also react with indole via priming of the early defensive signaling element [28], and when the plants were attacked by brown planthoppers, the elevated levels of guanine, inosine, hypoxanthine suggested that utilizing the purine salvage metabolism is a common adaptive process of rice plants to biotic stress [29]. The accumulation of primary and secondary metabolites is different depending on plant species, tissue type, developmental stage, and biotic stress conditions [30]. Whereas, in view of the particularity of the geographical location and species of *C. tetracocca*, alterations in metabolites concerning *E. onukii* piercing *C. tetracocca* are poorly known.

Metabolomics is a promising analytical tool for detecting primary and secondary metabolites in a given organism under biotic and abiotic stress, and elucidating these metabolic networks [31]. It is usually used to analyze plant–insect interactions, mutant characterization, phenotyping, and the identification of biomarkers [32,33]. Several recent studies have documented how metabolomics has been employed in the field of insect attacks on plants [34]. However, these researches only adopted gas chromatography–mass spectrometry (GC/MS) or liquid chromatography–mass spectrometry (LC/MS). Metabolite profiling with GC/MS and LC/MS has not previously been performed with regard to *E. onukii* attacking *C. tetracocca*. Therefore, we used GC/MS and LC/MS to parse the metabolite profiles of *E. onukii*-attacked *C. tetracocca*. Moreover, the published literature was employed to identify secondary metabolites with underlying insecticidal or repellent activity, as these metabolites perhaps can help us develop prevention and control agents against *E. onukii*, especially within the *C. tetracocca* protection area in southwest Guizhou.

2. Materials and Methods

2.1. Collection of C. tetracocca Leaves

For the experiment, *C. tetracocca* was cultivated at the state-owned Pubai Forest Farm, Pu'an (25°25′57″ N, 104°59′36″ E), Guizhou, where it biennially originates from seeds of a consistent genetic background. The identical *C. tetracocca* branch was selected, and the tea plant cutting and seedling raising were carried out from June to September 2020. After cuttage, the tea branch was maintained and watered. Light, temperature and other conditions were fully utilized to ensure the normal growth of tea seedlings in the natural environment. Adult *E. onukii* were collected from the *C. tetracocca* cultivar plantation in the field outside the experimental area. After propagation for four generations, third-instar nymphs with high activity were used in this experiment. These leafhoppers were starved for 2 h and fed on a new shoot of *C. tetracocca*; 35 cm × 50 cm insec-rearing cages were used for every separate plant to avoid interference from other pests. Two experimental groups, the leafhopper feeding (E1, E2, E3) and leafhopper not feeding (N1, N2, N3) groups, were

set up, and three biological replicates were harvested from each group. Every replicate contained 100 individuals of *E. onukii*, and they were supplemented with 10 individuals every other day. We investigated the situation of the tip of the leaf every day, and we ended the experiment when the blade tip was about to become fire-like; then, we collected 2–3 tea shoots with one bud and two leaves from each plant (Figure 1). The leaves were instantly packaged in a piece of aluminum foil and deposited in liquid nitrogen until sampling was completed. They were then preserved at −80 °C for subsequent metabolite detection.



Figure 1. The tea tree (*C. tetracocca*) was attacked by *E. onukii*. (**A**) *C. tetracocca* grown in the field habitat. (**B**) Tea seeds of *C. tetracocca*. (**C**) The sprout of *C. tetracocca* grown in the field habitat. (**D**) The leaves of *C. tetracocca* attacked by *E. onukii*. (**E**) The healthy leaves of *C. tetracocca*.

2.2. GC/MS Analysis

All the chemicals and solvents were HPLC grade. Water and methanol were purchased from Thermo Fisher Scientific (Thermo Fisher Scientific, Waltham, MA, USA). Pyridine, n-hexane, methoxylamine hydrochloride (97%), and BSTFA with 1% TMCS were purchased from CNW Technologies GmbH (Düsseldorf, Germany). Chloroform was obtained from Titan Chemical Reagent Co., Ltd. (Shanghai, China). L-2-chlorophenylalanine was obtained from Shanghai Heng-chuang Bio-technology Co., Ltd. (Shanghai, China).

An accurately weighed sample of 60 mg was placed in a 1.5 mL Eppendorf tube with two small steel balls and 600 µL of methanol–water (V:V = 1:1, containing 4 µg/mL of L-2-chlorophenylalanine) solution. Samples were kept at −40 °C for 2 min, and then they were ground at 60 HZ for 2 min using a full-automatic sample fast grinding instrument (Wonbio-E, Shanghai Wonbio Biotechnology Co., Ltd., Shanghai, China). All samples were ultrasonically treated by ultrasonic for 30 min in an ice-water bath. 150 µL chloroform was added to the samples, and the mixtures were vortexed for 2 min. The whole samples were extracted again using the same methods in ice-water bath, and then placed at −40 °C for 30 min. The mixture was centrifuged at 13,000× *g* rpm for 10 min at 4 °C. The supernatant (150 µL) was transferred in a glass vial and dried completely in a freeze concentration

centrifugal dryer. Subsequently, 80 μL of 15 mg/mL methoxylamine hydrochloride in pyridine was added. The oximation reaction was performed at 60 °C for 37 min after vortex oscillation for 2 min. The samples were derivatized with 50 μL of N, O-bis (trimethylsilyl) trifluoroacetamide (BSTFA) reagent and 20 μL of n-hexane with ten internal standard solutions (C8/C9/C10/C12/C14/C16/C18/C20/C22/C24, chloroform) 10 μL at 70 °C for 60 min. Quality control (QC) samples, i.e., mixtures of all samples to be analyzed, were mixed with an aliquot of all the samples to be utilized through the process of detection.

The derivatized samples were measured and analyzed on an Agilent 7890B gas chromatography system coupled with an Agilent 5977B MSD system (Agilent Technologies Inc., Santa Clara, CA, USA). A DB-5MS fused-silica capillary column (30 m × 0.25 mm × 0.25 μm, Agilent J & W Scientific, Folsom, CA, USA) was adopted to seperate the derivatives. The processes employed helium (>99.999%) as the carrier gas with a stable flow rate of 1 mL/min. The injector temperature was maintained at 260 °C, and injection volume was 1 μL in splitless mode. The initial oven temperature was held at 60 °C for 0.5 min; was ramped up to 125 °C at a rate of 8 °C/min, reached to 210 °C at a rate of 4 °C/min, raised to 270 °C at a rate of 5 °C/min, and to 305 °C at a rate of 10 °C/min; and finally, was retained at 305 °C for 3 min. The temperatures of the MS quadrupole and ion source were set to 150 and 230 °C, respectively. The collision energy was 70 eV. Mass data were acquired in full-scan mode (m/z 50–500), and the solvent delay time was set to 5 min.

2.3. LC/MS Analysis

All the chemicals and solvents were HPLC grade. Water, methanol, acetonitrile and formic acid were purchased from Thermo Fisher Scientific (Thermo Fisher Scientific, Waltham, MA, USA). L-2-chlorophenylalanine was obtained from Shanghai Heng-chuang Bio-technology Co., Ltd. (Shanghai, China).

An accurately weighed sample of 60 mg was transferred and deposited into a 1.5 mL Eppendorf tube with the addition of two small steel balls. A total amount of 600 μL of methanol–water (V:V = 7:3) was added to the mixture consisting of 4 μg/mL L-2-chlorophenylalanine. The samples were kept at −20 °C for 2 min, then were withdrawn to be ground at 60 HZ for 2 min, and were ultrasonically extracted for 30 min in an ice-water bath.

Next, all the mixtures were placed at −40 °C overnight. Samples were centrifuged at 13,000× *g* rpm for 10 min at 4 °C. Then 150 μL of supernatants was collected and filtered by 0.22 μm microfilters. The supernatant was transferred to a fresh 2 mL LC/MS glass vial, and stored at −80 °C until LC-MS analysis. QC was prepared by mixing an aliquot of all samples to creat a pooled sample.

The ACQUITY UPLC I-Class system (Waters Corporation, Milford, CT, USA) coupled with the VION IMS QTOF mass spectrometer (Waters Corporation, Milford, CT, USA) was employed to parse the metabolic profiling in both ESI-positive and ESI-negative ion modes. The system used an ACQUITY UPLC BEH C18 column (1.7 μm, 2.1 × 100 mm) in both positive and negative modes. The mobile phase A was water with 0.1% formic acid and the mobile phase B was acetonitrile with 0.1% formic acid. The linear gradient was as follows: 0.01 min, 5% B; 2 min, 5% B; 4 min, 30% B; 8 min, 50% B; 10 min, 80% B; 14 min, 100% B; 15 min, 100% B; 15.1 min 5% B; and 16 min, 5% B. The flow rate was 0.35 mL/min and column temperature was 45 °C. All the samples were maintained at 4 °C during the analysis. The injection volume was 1 μL. Data acquisition was performed in full scan mode with an m/z ranges from 50 to 1000 combined with MSE mode, consisting of two independent scans with different collision energies (CEs) that were alternatively acquired during the run. Parameters of mass spectrometry were set on two modes to fragment the ions, including a low-energy scan (CE 4 eV), and a high-energy scan (CE ramp 20–45 eV). Argon (99.999%) was employed as a collision-induced dissociation gas. ESI source conditions were set according to the conditions: scan time—0.2 s; interscan delay—0.02 s; capillary voltage—2.5 kV; cone voltage—40 V; source temperature—115 °C; desolvation gas temperature—450 °C; desolvation gas flow—900 L/h.

2.4. Data Preprocessing and Annotation

For GC/MS, the acquired GC/MS raw data (.D format) were shifted to the .abf format through software Analysis Base File Converter software for quick retrieval of data. Then, data were introduced into the MS-DIAL software, which performs peak detection, peak identification, MS2Dec deconvolution, characterization, peak alignment, wave filtering, and missing value interpolation.

For LC/MS, the original LC-MS data were processed via Progenesis QI V2.3 software (Nonlinear, Dynamics, Newcastle, UK) for baseline filtering, peak identification, integration, retention time correction, peak alignment, and normalization. A 5 ppm precursor tolerance, 10 ppm product tolerance, and 5% product ion threshold were applied.

2.5. Compound Identification

For GC/MS, a three-dimensional matrix was derived from the LUG database, which consists of the sample information, name of the peak of each substance, retention time, retention index, mass-to-charge ratio, and signal intensity. Internal standard peak and false positive peak in the original data matrix, including noise, column loss and derivative reagent peak were removed, and the ion peak with missing values of each group (ion intensity = 0) higher than 50% were deleted. The remaining missing value (ion intensity = 0) was replaced by half of the minimum value. All peak signal intensities (peak areas) are normalized by segments according to the internal standard with filtered RSD less than 0.1. After the data is normalized, the redundancy and peak merging are carried out, and the qualitative compounds are screened according to the total score of the qualitative results of the compounds. Compounds with resulting scores below 70 (out of 100) points were also deemed to be inaccurate and were removed.

For LC/MS, compound identification was conducted on the basis of a meticulous mass-to-charge ratio (m/z), secondary fragments, and isotopic distribution via The Human Metabolome Database (HMDB), Lipidmaps (V2.3), Metlin, EMDB, PMDB, and self-built databases for qualitative analysis. The extracted data were then subsequently processed by removing any peaks with a missing value (ion intensity = 0) in more than 50% of groups by substituting the zero value with half of the minimum value, and by screening according to the qualitative results of the compound. Compounds with resulting scores below 36 (out of 60) points were also deemed to be inaccurate and were removed. A data matrix was combined from the positive and negative ion data.

2.6. Data Analysis

After the unitariness of the data was achieved, redundancy elimination and peak merging were performed to obtain the data matrix. The data matrix of both GC/MS and LC/MS was inputted to the R package to execute the principle component analysis (PCA), which can investigate the overall distribution among the samples and the stability of the whole analysis process. Orthogonal partial least-squares-discriminant analysis (OPLS-DA) and partial least-squares-discriminant analysis (PLS-DA) were applied to differentiate the metabolites that differ between groups. In order to prevent overfitting, a 7-fold cross-validation and 200-response permutation testing (RPT) were used to measure the quality of the model. Variable importance of projection (VIP) values obtained from the OPLS-DA model were used to rank the overall contribution of each variable to group discrimination. A two-tailed Student's T-test was further adopted to prove whether the difference in metabolites between groups was significant. Differential metabolites were selected with VIP values greater than 1.0 and p-values less than 0.05 in MetaboAnalyst 4.0. Then, by summarizing the metabolites of damaged leaves and reviewing related references, we utilized ChemDraw to visualize series of compounds related to insect resistance.

3. Results

3.1. Metabolite Identification for Leaves of C. tetracocca

The GC/MS and LC/MS platforms, together with annotation software and databases, were chose to identify metabolites from healthy and damaged leaves of *C. tetracocca* attacked by *E. onukii*. Differences in metabolites between damaged and disease-free leaves were assessed by gauging three biological replicates. The GC-MS chromatograms of six samples from damaged and healthy leaves displayed good repeatability, proving that the operating conditions were stable and reliable (Figure 2). The retention times and peak areas of three QC samples also displayed good repeatability during the experiment (Supplementary Figure S1), signifying that the apparatus was extremely invariable. The relative deviation of the internal standard (3,4-dichlorophenylalanine and saturated fatty acid methyl ester) added to the QC sample validated the system's stability. In total, 56 metabolites were preliminarily identified via a mass spectrum match and retention index match. Primary mass spectrum of L-phenylalanine is shown in Figure 3A.

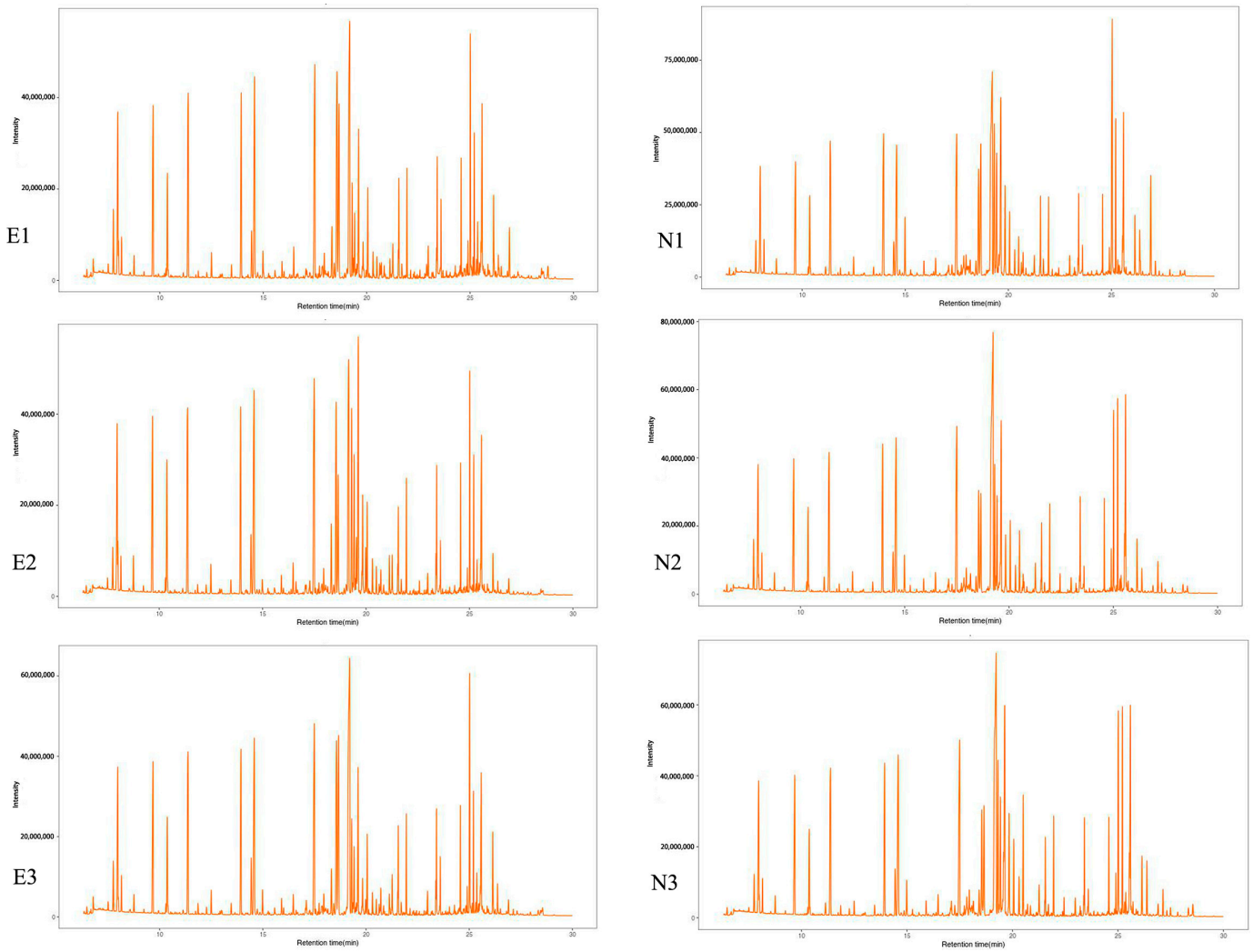


Figure 2. Total ion current (TIC) chromatogram of damaged leaves attacked by *E. onukii* of *C. tetracocca* and healthy leaves based on GC-MS. E1, E2 and E3 represents the groups of the leafhopper feeding, N1, N2 and N3 represents the groups of leafhopper not feeding.

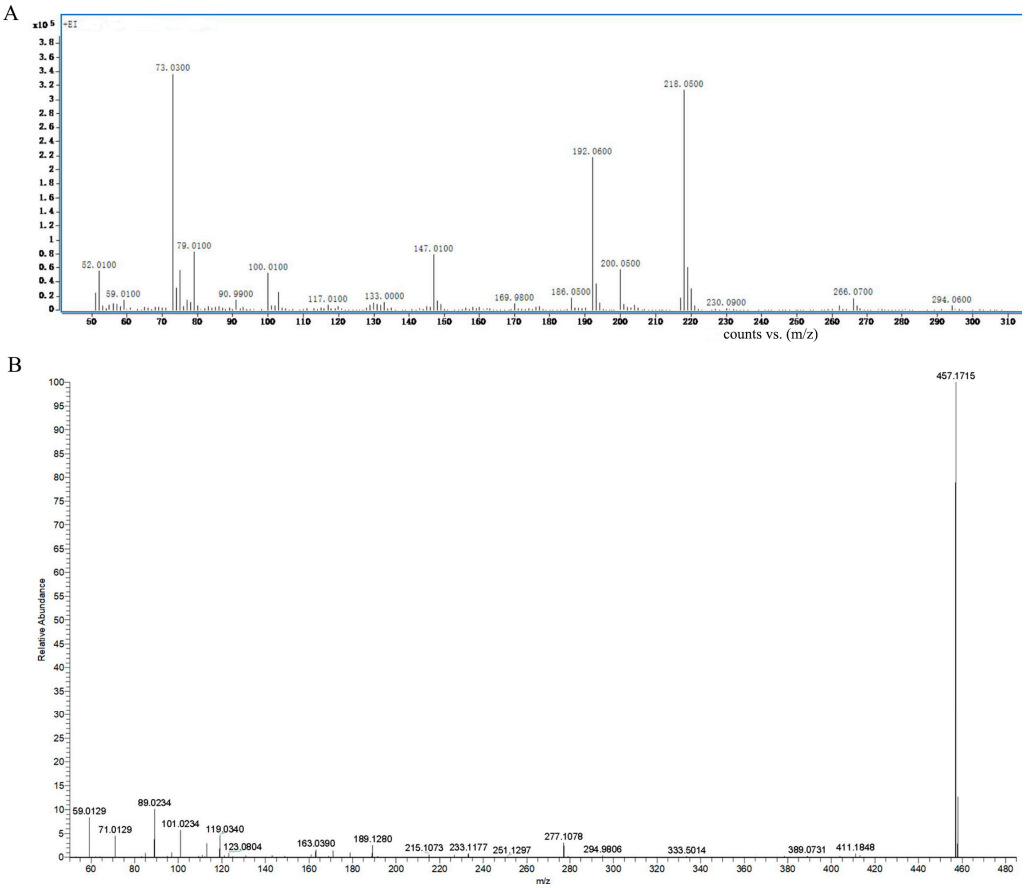


Figure 3. Primary mass spectrum of L-phenylalanine damaged leaves attacked by *E. onukii* of *C. tetracocca* (**A**). The secondary mass spectrum of compound epigallocatechin-7-O-gallate damaged leaves attacked by *E. onukii* of *C. tetracocca* (**B**).

Samples from damaged and healthy leaves were also assayed by the LC-MS platform, and total ion chromatograms (TICs) were obtained in positive and negative ion modes (Supplementary Figures S2 and S3). Three QC samples also showed good repetitiveness during the experiment (Supplementary Figure S4). The relative deviations of the internal standard (L-2-chlorophenylalanine) in the QC samples in the positive and negative ion modes also demonstrated that the instrument was extraordinarily stable. In total, 567 metabolites were presumptively identified in the positive and negative ion modes. The secondary mass spectrum of compound epigallocatechin-7-O-gallate is shown in Figure 3B.

3.2. Principal Component Analysis (PCA) and Orthogonal Projections to Latent Structures-Discriminant Analysis (OPLS-DA)

The similarities and differences between leaves damaged by *E. onukii* and the healthy leaves of *C. tetracocca* were summarized by using the principal component analysis (PCA). As for the GC/MS data, the PCA score plot revealed that the first four components explained 82.7% of the total variation in metabolite levels (Supplementary Figure S5); PC1 accounted for 47.6%, and PC2 accounted for 19.1% of the total variation (Figure 4A). The orthogonal projections to latent structures discriminant analysis (OPLS-DA) was adopted to eliminate the noise irrelevant to the classification information and to improve the analytical ability and effectiveness of the model. The model can distinguish between two experimental treatment groups (Figure 4B).

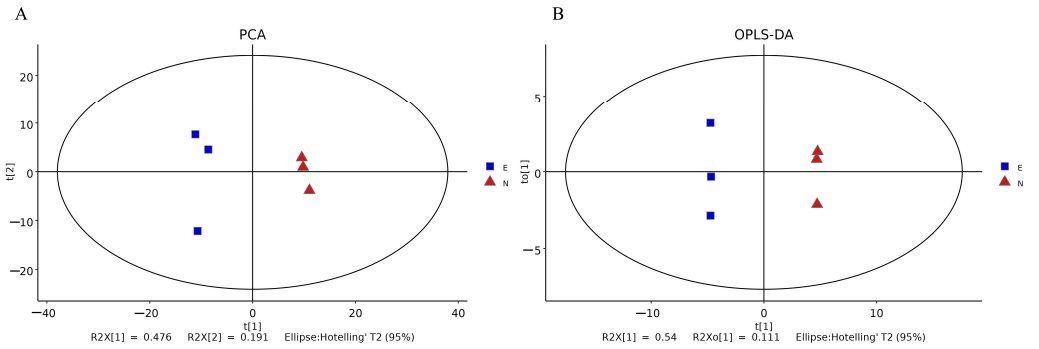


Figure 4. Data analysis of the metabolites based on GC/MS in the damaged leaves (E) of *E. onukii* and healthy leaves (N). (**A**) Principal component analysis (PCA). (**B**) Orthogonal projections to latent structures discriminant analysis (OPLS-DA).

As for the LC/MS data, the first four principal components explained 78.9% of the total variation in metabolites detected in both positive and negative ion modes (Supplementary Figure S6). In the score plot, PC1 accounted for 46.8% and PC2 accounted for 18.2% of the total variation in positive and negative ion modes (Figure 5A). Similarly, the herbivore-attacked leaves and healthy leaves of *C. tetracocca* were clearly distinguished on the OPLS-DA plots (Figure 5B). These results revealed that a notable difference existed in the metabolites of the damaged and healthy leaves of *C. tetracocca*.

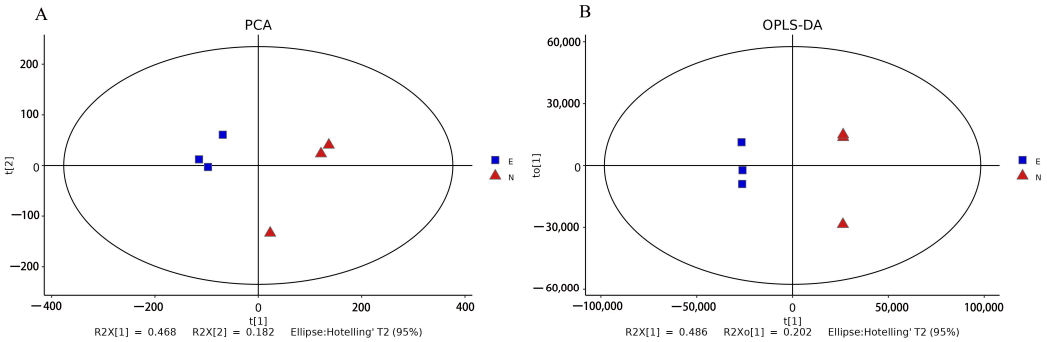
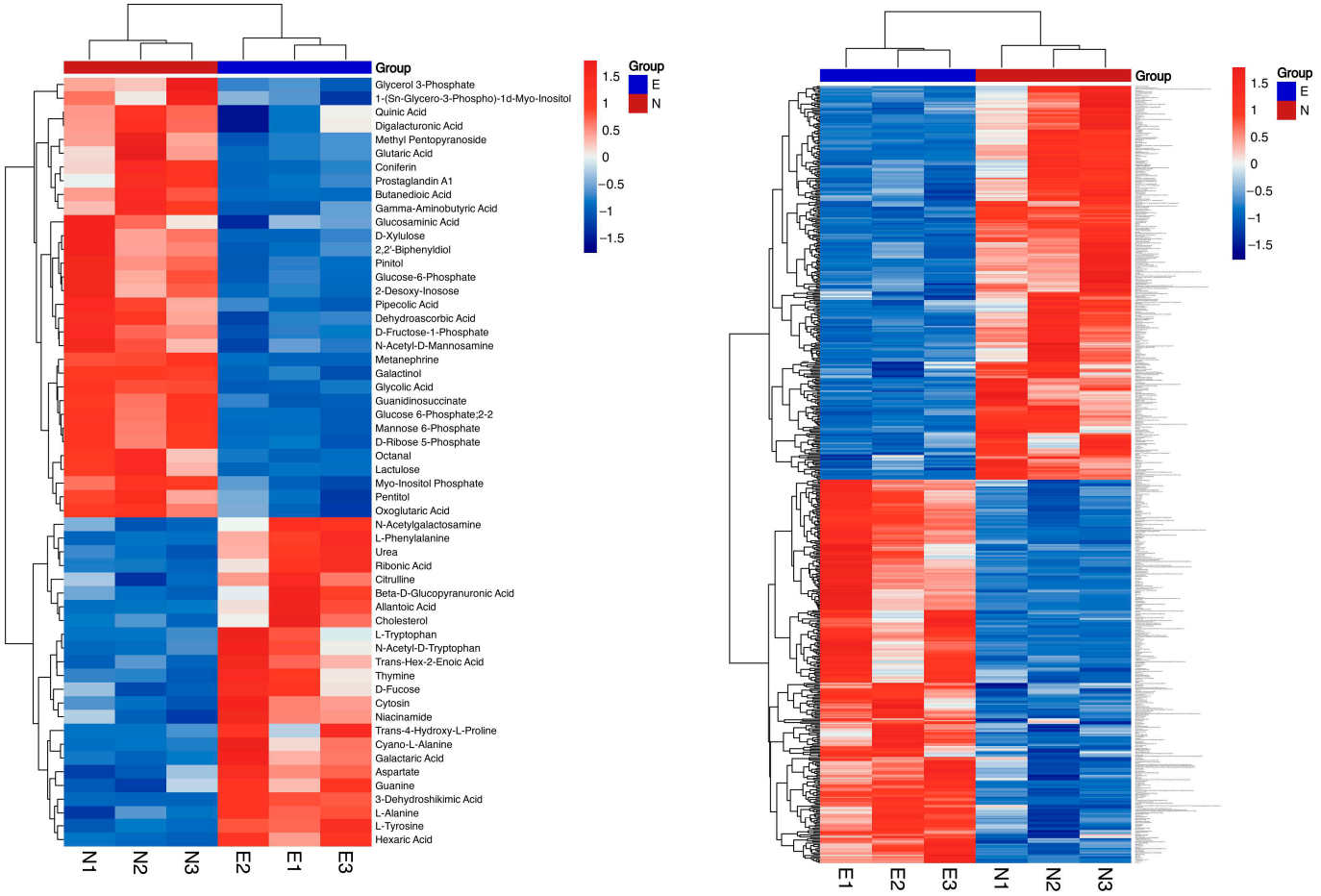


Figure 5. The data analysis of the metabolites based on LC/MS in the damaged leaves (E) of *E. onukii* and healthy leaves (N). (**A**) Principal component analysis (PCA) analysis in positive ion modes. (**B**) Orthogonal projections to latent structures discriminant analysis (OPLS-DA).

3.3. Differentially Abundant Metabolites and Metabolic Pathway Analysis

A Student's *t*-test and a fold-change analysis (fold-change > 1.0, $p < 0.05$) were used to screen for metabolites with a different abundance in MetaboAnalyst 4.0. Discrepancies in metabolite levels between two leaf types were exhibited by a heatmap. In total, 56 differentially abundant metabolites (24 up-regulated and 32 down-regulated) were identified in the leaves of *C. tetracocca* utilizing GC/MS (Figure 6A, Supplementary Table S1). Using LC/MS, 567 (280 up-regulated and 287 down-regulated) differentially abundant metabolites were acquired in both positive and negative ion modes (Figure 6B, Supplementary Table S2). In addition, we conducted a joint clustering of all differential metabolites of LC-MS and GC-MS (Figure 7). This result is consistent with the results of separate analysis of the two methods. Cluster analysis of the top 50 differential metabolites revealed that 10 compounds (indoleacrylic acid, shikimic acid, 8-O-acetyl shanzhiside methyl ester, medecassic acid, cimimanol D, coroloside, quercetin 3-O-rhamnosyl-(1->2)-rhamnosyl-(1->6)-glucoside), skimmianine, aginoside progenin, cis-3-hexenyl b-primeveroside, and theogallin) showed high levels on the leaves attacked by leafhoppers (Figure 8).



(**A**) (**B**)

Figure 6. Hierarchical clustering of the differentially abundant metabolites based on GC-MS (**A**) and both the positive and negative ion modes based on LC-MS (**B**) between healthy leaves (N1, N2, N3) and damaged leaves of *E. onukii* (E1, E2, E3). The data were log2 transformed, and similarity assessment for clustering was based on the Euclidean distance coefficient and complete clustering algorithm. Columns and rows represent individual metabolites and different samples respectively.

With the purpose of obtaining the most dependent metabolic pathways concerned with herbivorous insect resistance, the differentially abundant metabolites were mapped to the Kyoto Encyclopedia of Genes and Genomes (KEGG) database. All differential metabolites were annotated to the KEGG database (Figure 9). Based on GC-MS, these metabolites are mainly involved in phenylalanine, tyrosine and tryptophan biosynthesis; arginine biosynthesis and alanine; aspartate and glutamate metabolism. Based on LC-MS, these metabolites are mainly involved in flavonoid biosynthesis; ascorbate and aldarate metabolism; stilbenoid, diarylheptanoid and gingerol biosynthesis (Figure 9).

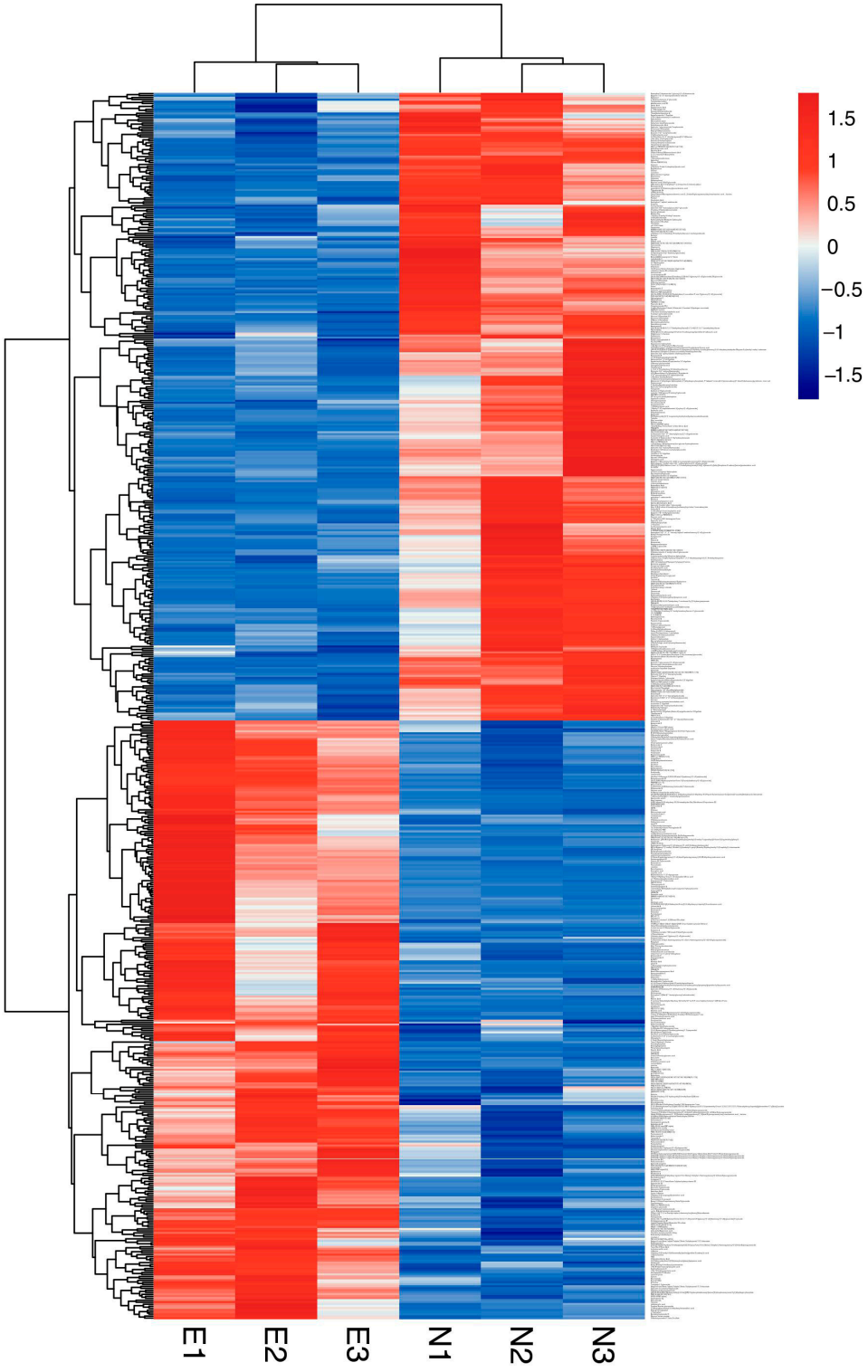


Figure 7. Hierarchical clustering of the differentially abundant metabolites based on combination with GC-MS and LC-MS. The data were log2 transformed, and similarity assessment for clustering was based on the Euclidean distance coefficient and complete clustering algorithm. Columns and rows represent individual metabolites and different samples respectively. E1, E2 and E3 represents the groups of the leafhopper feeding, N1, N2 and N3 represents the groups of leafhopper not feeding.

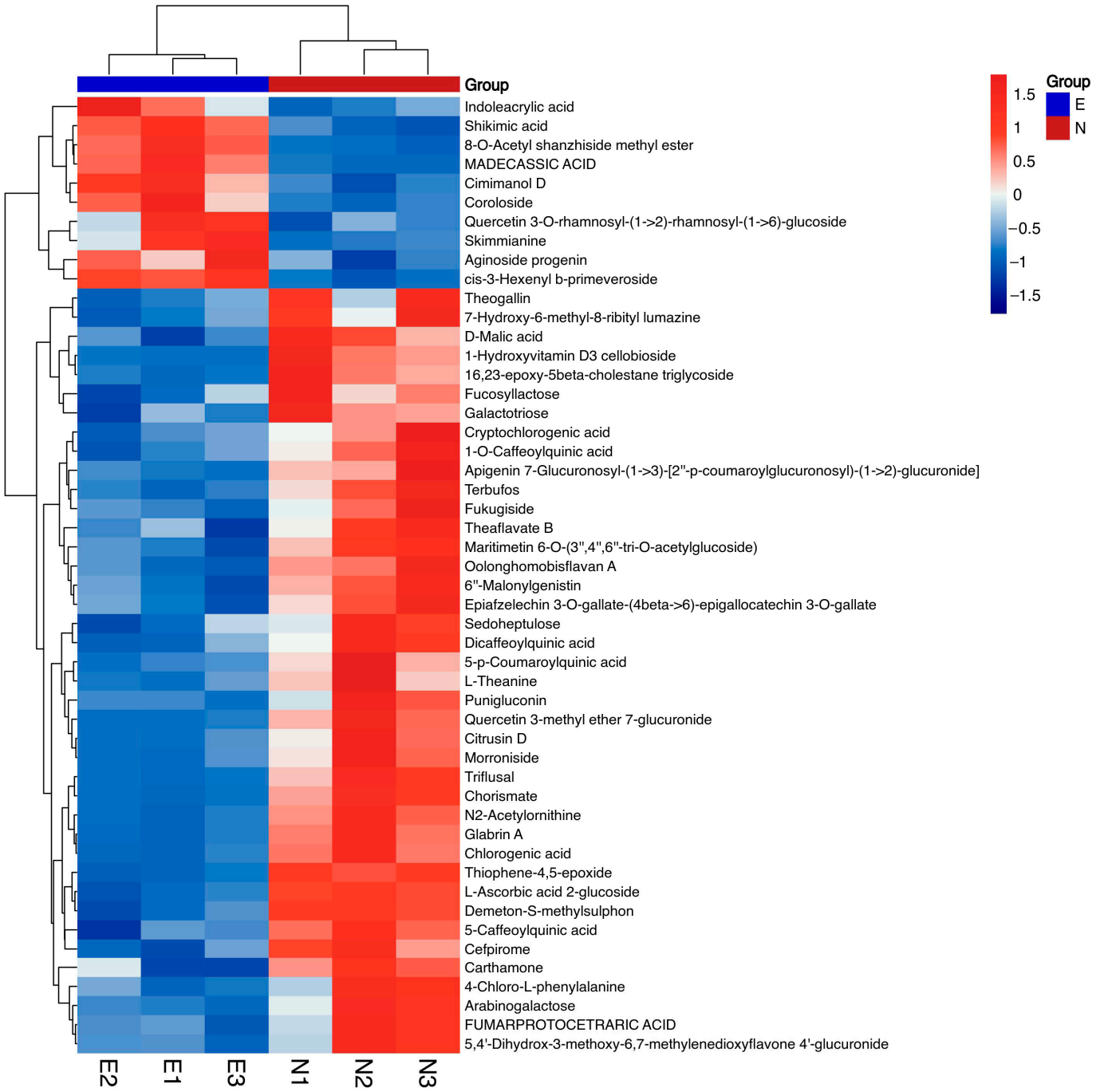


Figure 8. The hierarchical clustering of the top 50 differentially abundant metabolites based on both the positive ion modes and negative ion modes based on LC-MS in the healthy leaves and damaged leaves of *E. onukii*. The data were log2 transformed, and similarity assessment for clustering was based on the Euclidean distance coefficient and complete clustering algorithm. Columns and rows represent individual metabolites and different samples, respectively.

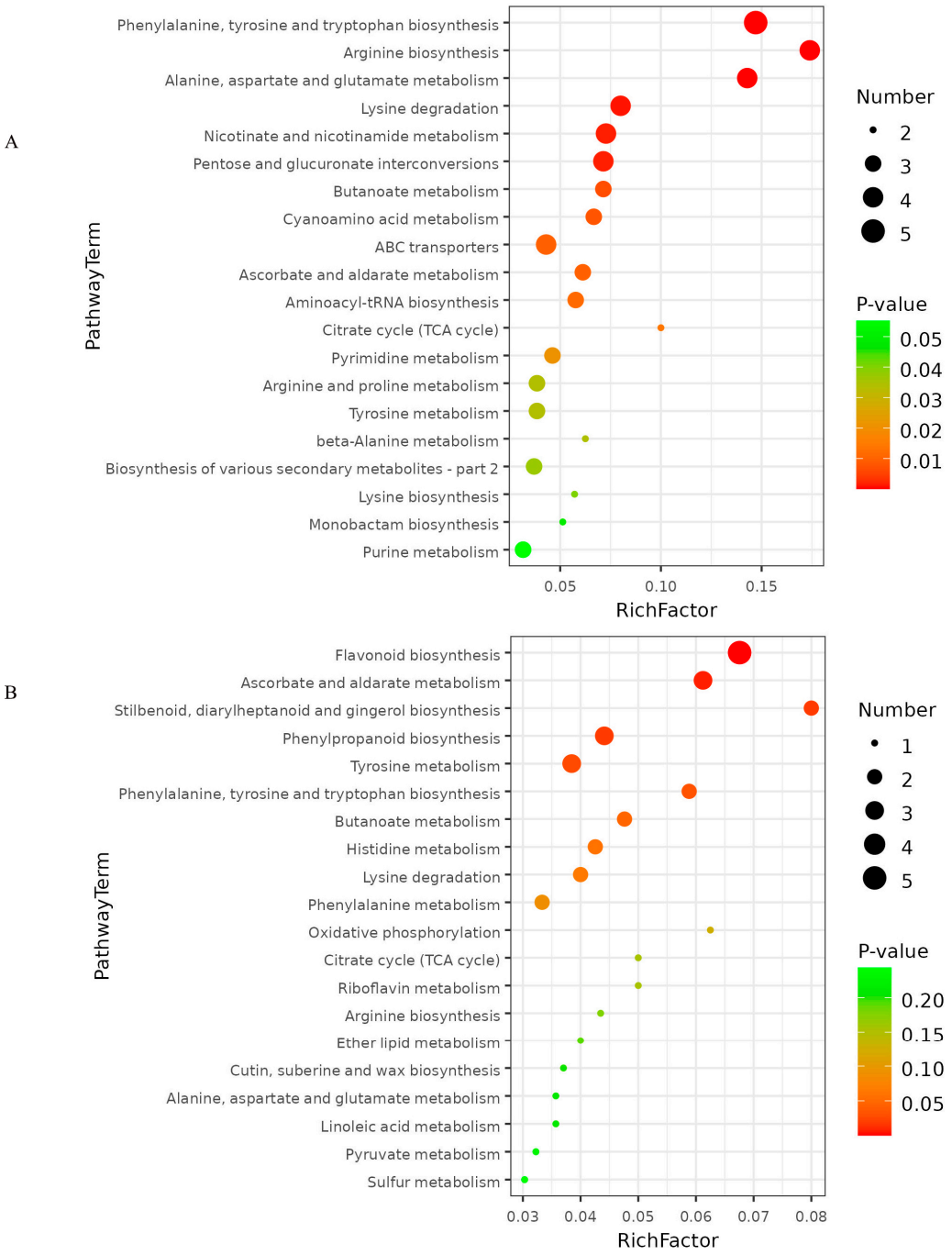


Figure 9. Pathway analysis of differentially abundant metabolites from GC-MS (**A**) and LC-MS (**B**) based on pathway-matched metabolites.

The top 20 metabolic pathways related to the differential metabolites were identified based on the combination with GC-MS and LC-MS. Its metabolic pathway mainly involves phenylalanine, tyrosine and tryptophan biosynthesis, arginine biosyhthesis; pentose and glucuronate interconversions; phenylpropanoid biosynthesis; alanine, aspartate and glutamate metabolism; ascorbate and aldarate metabolism; citrate cycle, Butanoate metabolism; Histidine metabolism; flavonoid biosynthesis (Figure 10). The results of these joint analyses are consistent with those of the two methods.

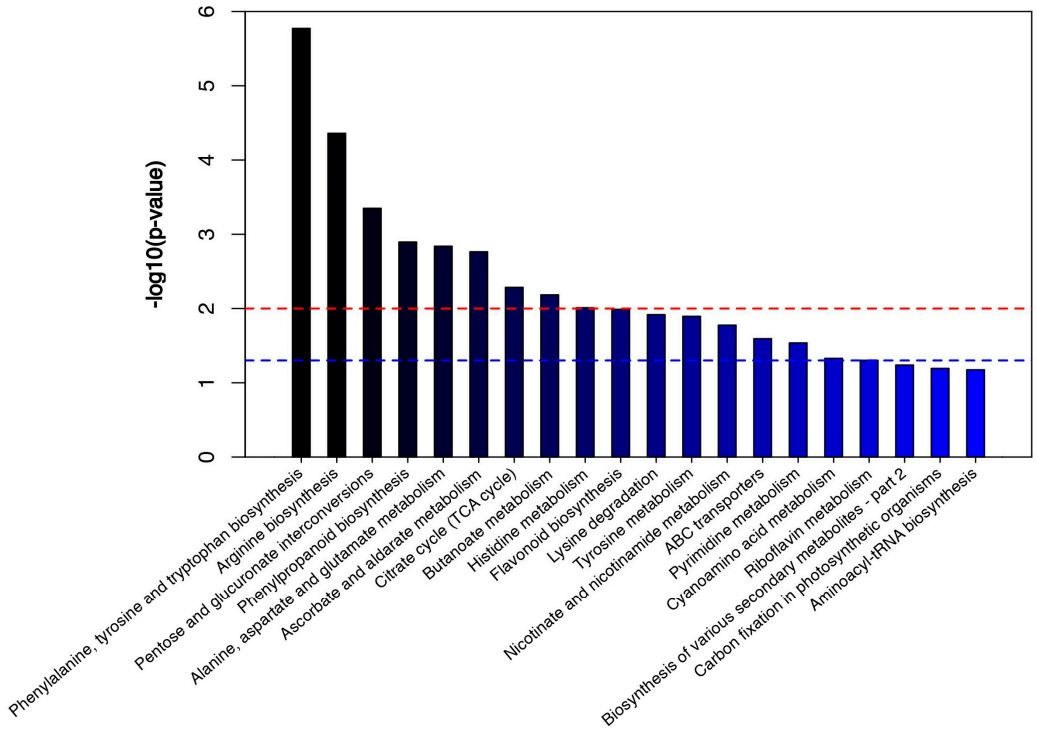


Figure 10. Enrichment of metabolic pathway were analysed by conbination with GC-MS and LC-MS. Bars that exceed the blue and red dotted lines indicate pathways with $p < 0.05$ and $p < 0.01$, respectively.

4. Discussion

In an attempt to battle the destructive behaviour of herbivorous insects, many plants produced defense compounds, nowadays called specialized metabolites [35,36]. They branch off from the primary metabolism involving amino acids, fatty acids, and hormones, which has ultimately created precursor pathways that are rather conserved among plant species [37]. Primary metabolites actively participate in the regulation of the regular growth and development of plants [38]. In addition to their direct engagement in plant growth, development, and reproduction, plant primary metabolites are also involved in insect defense reactions [39,40]. In this work, seven amino acids, peptides, and analogues including allantoic acid, L-phenylalanine, L-tyrosine, citrulline, *trans*-4-hydroxy-L-proline, aspartate, and L-alanine (Figure 6) were up-regulated in leaves of *C. tetracocca* attacked by *E. onukii*. Amino acids, the principal form of nitrogen that exists in plants, limit the growth of insect herbivores and are considered to be precursors for crowd defense-related plant metabolites [41]. Therefore, the regulation of amino acid biosynthesis is regarded as an important pathway to control tea green leafhopper's feeding. In addition, herbivore-induced changes in amino acid metabolism were mainly affected by insect attacks. In many plant species, the phloem transport of nitrogen is dominated by four amino acids, glutamine, glutamic acid, asparagine, aspartic acid [42]. A similar scenario was documented in other plants, where the amino acids of Japanese rowan (*Sorbus commixta*) leaves infested with apple-grass aphids (*Rhopalosiphum insertum*) were significantly up-regulated compared to uninfected leaves [43]. Chlorosis-inducing greenbugs (*Schizaphis graminum*) raise the amino acid contents in the phloem sap of wheat [44,45]. Furthermore, D-fucose was up-regulated, whereas lactulose and D-xylulose were down-regulated in attacked leaves (Figures 6 and 7). In this study, many organic acid and derivatives, including butanedioic acid, oxoglutaric acid and glycolic acid were present at higher levels in normal leaves (Figures 6 and 7).

A crowd of specialized metabolites present multiple biological functions in protecting plants against abiotic and biotic stresses to adapt to the variable environment [46]. In this study, 567 differentially abundant metabolites were discovered: of these, 280 metabolites were up-regulated and 287 metabolites were down-regulated (Figure 6). Based on the

previous literature reports, catechins, containing catechin (C), epicatechin (EC), gallocatechin (GC), epicatechin-3-gallate (ECG), epigallocatechin (EGC), and epigallocatechin-3-gallate (EGCG), caffeine, volatile compounds were related to the resistance against insect attacks [47,48]. However, gallocatechin 3′-gallate, epigallocatechin 7-O-gallate, and gallocatechin 7,4′-di-O-gallate were present at low levels in the leaves attacked by the leafhopper. The differential metabolites mainly composed of amino acids, flavonoids, terpenoids, terpene glycosides, organooxygen compounds, fatty acyls, polyketides, glycerophospholipids and steroids. Most of the flavonoids were down-regulated, whereas terpenoids (16beta-16-hydroxy-3-oxo-1,12-oleanadien-28-oic acid, 23-hydroxybetulinic acid, 3beta,15alpha-diacetoxylanosta-8,24-dien-26-oic acid, centellasapogenol A, epoxyganoderiol C, euscaphic acid, ganolucidic acid C, liquiritic acid, madlongiside D, medicoside C, notoginsenoside R1, phytolaccinic acid, sandosapogenol, theasapogenol E, and *trans*-3-feruloylcorosolic acid) and terpene glycosides (araliasaponin I, calenduloside H methyl ester, cinncassiol C1 19-glucoside, diosbulbinoside F, lasalocid, metabolite A, momordcin I, soyasaponin bg, soyasaponin gamma-G, suspensolide F, and tsangane L 3-glucoside) were up-regulated. This phenomenon is different from that which occurs when *E. onukii* attacks *Camellia sinensis* [49]. This result once again proves that there is a unambiguous difference between local tea (*C. tetracocca*) and *C. sinensis*. Interestingly, all terpenoids are concentrated in leafhopper-attacked leaves without exception.

Cluster analysis of the top 50 differential metabolites revealed that ten compounds showed high levels on the leaves attacked by tea green leafhoppers (Figure 8). Shikimic acid is the precursor of the shikimate pathway, and it is produced from a combination of erythrose 4-phosphate (pentose phosphate pathway) and phosphoenolpyruvate (glycolytic pathway). The amino acids phenylalanine, tyrosine and tryptophan are the products of the shikimate pathway, and are deemed as precursors of phenolics and nitrogen-containing secondary metabolites [50].

We summarized the metabolites of damaged leaves attacked by leafhoppers based on analysis of experimental results and review of related documents (Figure 11). The two main compounds are momordicin I and arabidopside B [51]. Momordicin I showed conspicuous antifeedant activity on the larvae of *Plutella xylostella*, where its concentrations for AFC_{50} against the second and the third instar were 144.08 and 168.42 $\mu g \cdot mL^{-1}$ [52]. Momordicin I may also be one of the metabolites of the tea green leafhopper. Arabidopsides, as one of esterified oxylipins, portray a crucial role in plant defense mechanisms. *Arabidopsis thaliana* esterified oxylipins peaks under different extensive biotic stresses. Arabidopsides B was discovered through wounding of *Arabidopsis arenosa* (L.) [53]. Then, these metabolites were detected in two other Arabidopsis species and three other species of the Brassicaceae [54], which confirmed that these molecules are not limited to the genus Arabidopsis. Our results conform to the mechanism of *Arabidopsis* resistance. Overall, momordicin I and arabidopside B can be used as a potential biological control agents.

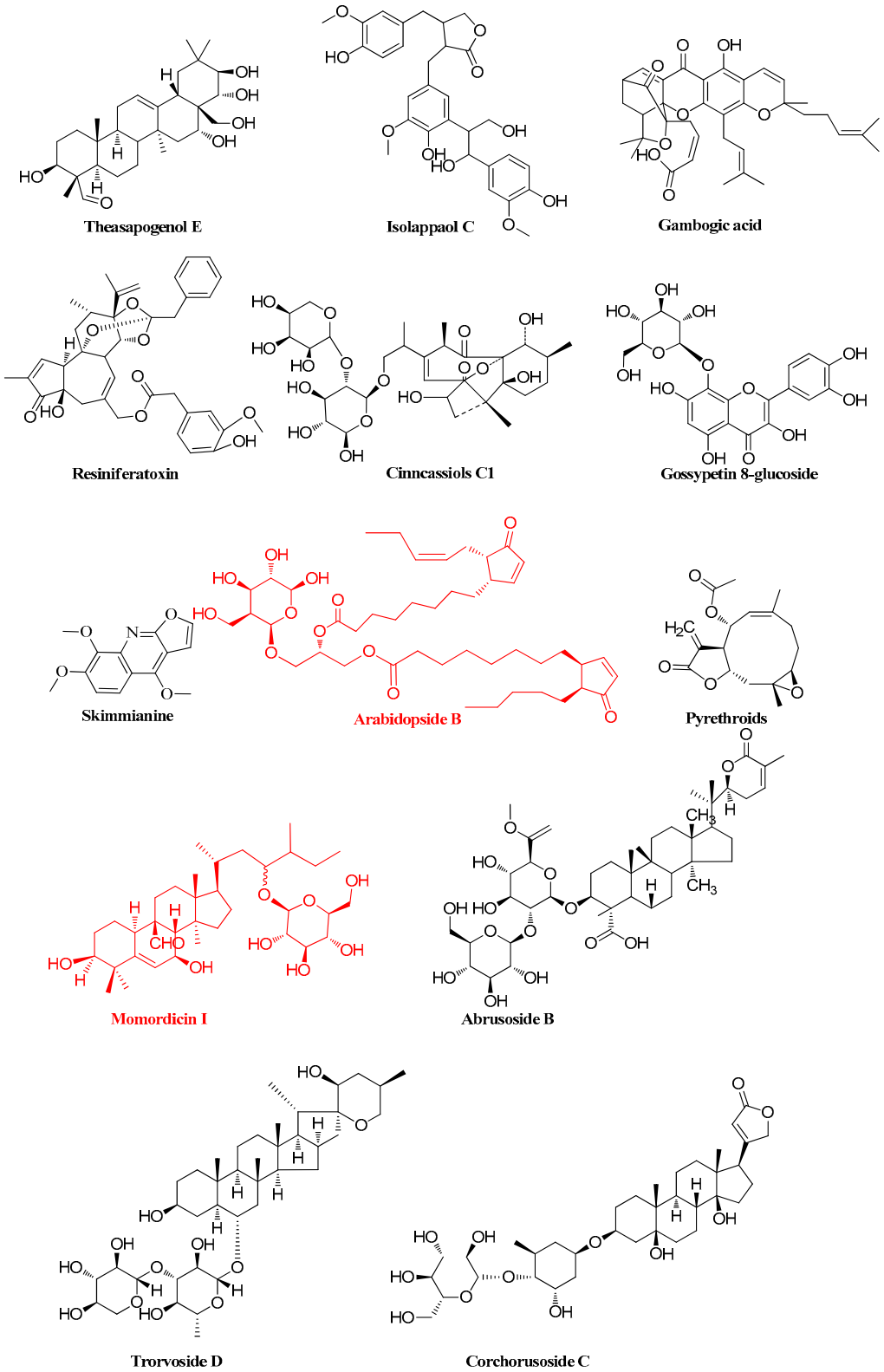


Figure 11. 14 Metabolites related to insect resistances based on up-regulated compounds derived from the after *E. onukii* feeding the *C. tetracocca* and summary of relevant literatures, red represent the most potential insect-resistant metabolite.

5. Conclusions

We compared the metabolite profiles of leaves of *C. tetracocca* attacked by *E. onukii* and healthy leaves collected during the experiment. The accumulation of many specialized

metabolites in *C. tetracocca* leaves establishes that they were induced by the insect attacked. Triterpenoids have the potential for development as biocontrol agents for *C. tetracocca*. In future research, we will measure and test the repellent activity of these metabolites in leaves of *C. tetracocca* attacked by leafhoppers. Meanwhile, two compounds (momordicin I and arabidopside B), will be the focus on when screening compounds for the prevention and control of *E. onukii*. The results will lay a foundation for the pest control of *C. tetracocca* in the Pubai Forest Farm.

Supplementary Materials: The following supporting information can be downloaded at: https://www.mdpi.com/article/10.3390/agronomy13030791/s1, Figure S1: Total ion current (TIC) chromatogram of QC Samples in GC-MS; Figure S2: Total ion current (TIC) chromatogram of negative ion in LC-MS Samples; Figure S3: Total ion current (TIC) chromatogram of positive ion in LC-MS samples; Figure S4: Total ion current (TIC) chromatogram of negative (A) and positive ion (B) of QC samples in LC-MS; Figure S5: PCA score plot (PC1, PC2, PC3) in GC-MS. Blue squares represent group E, red circles represent group N; Figure S6: PCA score plot (PC1, PC2, PC3) in GC-MS. Blue squares represent group E, red circles represent group N. Table S1: All differential metabolites based on GC-MS; Table S2: All differential metabolites based on LC-MS.

Author Contributions: Conceptualization, N.Z. and Y.S.; methodology, X.Z.; software, T.P.; validation, W.T. and J.W.; formal analysis, G.L.; investigation, X.Z.; resources, N.Z.; data curation, J.W.; writing—original draft preparation, N.Z.; writing—review and editing, Y.S.; visualization, W.T.; supervision, Y.S.; project administration, Y.S.; funding acquisition, Y.S. All authors have read and agreed to the published version of the manuscript.

Funding: This research was funded by the World Top Discipline Program of Guizhou Province: Karst Ecoenvironment Sciences (No.125 2019 Qianjiao Keyan Fa), the Science and Technology Project of Guiyang City ([2020]7-18), the Innovation Group Project of Education Department of Guizhou Province ([2021]013), the National Natural Science Foundation of China (32260120) and the Natural Science Foundation of Guizhou Province (Qiankehejichu-ZK [2023]General 257). The APC was funded by 32260120.

Data Availability Statement: All data supporting the results of this study are included in the manuscript and data sets area vailable upon request.

Conflicts of Interest: The authors declare no conflict of interest.

References

1. Zhang, H.D. Systematic classification of tea plants. *Acta Sci. Nat. Univ. Sunyatseni* **1981**, *1*, 87–99.
2. Luo, W.; Chen, Y.Y.; Yin, S.H.; Wang, L.F.; Huang, X.X.; Geng, F.; Chen, X.M. Genetic diversity of *Camellia tetracocca* Zhang germplasm resources in Pu'an, Guizhou Province. *Acta Agric. Univ. Jiangxiensis* **2022**, *44*, 74–85.
3. Yuan, M.Q.; Qian, C.J. Study on endemic plant Camellia tetracocca in Pu'an.an county, Guizhou. *Guizhou Sci.* **2009**, *2*, 80–85.
4. Duan, X.Y.; Hu, H.J.; Wang, J.L.; Cao, Y.; Zhao, H.F. Present situation, protection and utilization of big tea tree Resources in southwest Guizhou. *Agric. Tech. Serv.* **2011**, *28*, 1736–1737.
5. Li, Y.F.; Ouyang, S.H.; Chang, Y.Q.; Wang, T.M.; Li, W.X.; Tian, H.Y.; Cao, H.; Kurihara, H.; He, R.R. A comparative analysis of chemical compositions in *Camellia sinensis* var. Pu'anensis Kurihara, a novel Chinese tea, by HPLC and UFLC-Q-TOF-MS/MS. *Food Chem.* **2017**, *216*, 282–288. [CrossRef]
6. Liu, Y.J.; Zhao, G.F.; Li, X.; Shen, Q.; Wu, Q.; Zhuang, J.H.; Zhang, X.Q.; Xia, E.H.; Zhang, Z.Z.; Qian, Y.M.; et al. Comparative analysis of phenolic compound metabolism among tea plants in the section Thea of the genus *Camellia*. *Food Res. Int.* **2020**, *135*, 109276. [CrossRef] [PubMed]
7. Li, J.Y.; Shi, L.Q.; Vasseur, L.; Zhao, Q.; Chen, J.; You, M.S.; You, S.J. Genetic analyses reveal regional structure and demographic expansion of the predominant tea pest *Empoasca onukii* (Hemiptera: Cicadellidae) in China. *Pest Manag. Sci.* **2022**, *78*, 2838–2850. [CrossRef] [PubMed]
8. Mao, Y.; Tan, R.; Gong, Z.; Kuang, S. Relationships between population dynamics of *Empoasca vitis* and meteorological factors in tea plantation. *Hubei Agric. Sci.* **2014**, *53*, 32.
9. Chen, L.L.; Yuan, P.; Pozsgai, G.; Chen, P.; Zhu, H.P.; You, M.S. The impact of cover crops on the predatory mite Anystis baccarum (Acari, Anystidae) and the leafhopper pest *Empoasca onukii* (Hemiptera, Cicadellidae) in a tea plantation. *Pest Manag. Sci.* **2019**, *75*, 3371–3380. [CrossRef] [PubMed]
10. Qin, D.; Zhang, L.; Xiao, Q.; Dietrich, C.; Matsumura, M. Clarification of the identity of the tea green leafhopper based on morphological comparison between Chinese and Japanese specimens. *PLoS ONE* **2015**, *10*, e0139202. [CrossRef]

11. Yao, Q.; Wang, M.Q.; Chen, Z.M. The relative preference of *Empoasca onukii* (Hemiptera: Cicadellidae) for oviposition on twenty-four tea cultivars. *J. Econ. Entomol.* **2022**, *115*, 1521–1530. [CrossRef]
12. Kosugi, Y. Infuence of injury by tea green leafhopper, *Empoasca onukii* Matsuda on leaves in new shoots of tea plants. *Tea Res. Bull.* **2000**, *88*, 1–8. (In Japanese)
13. Backus, E.A.; Serrano, M.S.; Ranger, C.M. Mechanisms of hopperburn: An overview of insect taxonomy, behavior, and physiology. *Annu. Rev. Entomol.* **2005**, *50*, 125–151. [CrossRef] [PubMed]
14. Muday, G.K.; Brown-Harding, H. Nervous system-like signaling in plant defense. *Science* **2018**, *361*, 1068–1069. [CrossRef]
15. Bai, Y.C.; Yang, C.Q.; Halitschke, R.; Paetz, C.; Kessler, D.; Burkard, K.; Gaquerel, E.; Baldwin, L.T.; Li, D.P. Natural history—Guided omics reveals plant defensive chemistry against leafhopper pests. *Science* **2022**, *375*, eabm2948. [CrossRef] [PubMed]
16. Mitchell, C.; Brennan, R.M.; Graham, J.; Karley, A.J. Plant defense against herbivorous pests: Exploiting resistance and tolerance traits for sustainable crop protection. *Front. Plant Sci.* **2016**, *7*, 1132. [CrossRef]
17. Zhou, Q.Y.; Zhao, S.Q.; Zhu, J.Y.; Li, F.D.; Tong, W.; Liu, S.R.; Wei, C.L. Transcriptomic analyses reveal a systemic defense role of the uninfested adjacent leaf in tea plant (*Camellia sinensis*) attacked by tea geometrids (*Ectropis obliqua*). *Genomics* **2020**, *112*, 3658–3667. [CrossRef]
18. Khare, S.; Singh, N.B.; Singh, A.; Hussain, I.; Niharika, K.; Yadav, V.; Bano, C.; Yadav, R.K.; Amist, N. Plant secondary metabolites synthesis and their regulations under biotic and abiotic constraints. *J. Plant Biol.* **2020**, *63*, 203–216. [CrossRef]
19. War, A.R.; Paulraj, M.G.; Ahmad, T.; Buhroo, A.A.; Hussain, B.; Ignacimuthu, S.; Sharma, H.C. Mechanisms of plant defense against insect herbivores. *Plant Signal Behav.* **2012**, *7*, 1306–1320. [CrossRef]
20. Thakur, M.; Bhattacharya, S.; Khosla, P.K.; Puri, S. Improving production of plant secondary metabolites through biotic and abiotic elicitation. *J. Appl. Res. Med. Aroma. Plants* **2019**, *12*, 1–12. [CrossRef]
21. Schoonhoven, L.M.; van Loon, J.J.A.; Dicke, M. Plant chemistry: Endless variety. In *Insect-Plant Biology*; Oxford University Press: Oxford, UK, 2005; pp. 3–82.
22. Harborne, J.B.; Baxter, H. Phytochemical dictionary. In *A Handbook of Bioactive Compounds from Plants*; Taylor & Francis: London, UK, 1993.
23. Harborne, J.B. *Introduction to Ecological Biochemistry*, 4th ed.; Academic Press: London, UK, 1993.
24. Hrborne, J.B. Phenolics. In *Natural Products: Their Chemistry and Biological Significance*; Longman: Harlow, UK, 1998; pp. 362–388.
25. Bahmani, M.; Golshahi, H.; Saki, K.; Rafieian-Kopaei, M.; Delfan, B.; Mohammadi, T. Medicinal plants and secondary metabolites for diabetes mellitus control. *Asian Pac. J. Trop. Dis.* **2014**, *4*, S687–S692. [CrossRef]
26. Shi, J.H.; Liu, H.; Pham, T.C.; Hu, X.J.; Liu, L.; Wang, C.; Foba, C.F.; Wang, S.B.; Wang, M.Q. Volatiles and hormones mediated root-knot nematode induced wheat defense response to foliar herbivore aphid. *Sci. Total Environ.* **2022**, *815*, 152840. [CrossRef]
27. Romero, B.; Dillon, F.M.; Zavala, J.A. Different soybean cultivars respond differentially to damage in a herbivore-specific manner and decreas herbivore performance. *Arthropod-Plant Interact.* **2019**, *14*, 89–99. [CrossRef]
28. Ye, M.; Glauser, G.; Lou, Y.; Erb, M.; Hu, L. Molecular dissection of early defense signaling underlying volatile-mediated defense regulation and herbivore resistance in Rice. *Plant Cell* **2019**, *31*, 687–698. [CrossRef]
29. Uawisetwathana, U.; Jamboonsri, W.; Bamrungthai, J.; Jitthiang, P.; Nookaew, I.; Karoonuthaisiri, N. Metabolite profiles of brown planthopper-susceptible and resistant rice (*Oryza sativa*) varieties associated with infestation and mechanical stimuli. *Phytochemistry* **2022**, *194*, 113044. [CrossRef] [PubMed]
30. Sampaio, B.L.; Edrada-Ebel, R.; Costa, F.B.D. Effect of the environment on the secondary metabolic profile of Tithonia diversifolia: A model for environmental metabolomics of plants. *Sci. Rep.* **2016**, *6*, 29265. [CrossRef] [PubMed]
31. Kumar, R.; Bohra, A.; Pandey, A.K.; Pandey, M.K.; Kumar, A. Metabolomics for Plant Improvement: Status and Prospects. *Front. Plant Sci.* **2017**, *8*, 1302. [CrossRef] [PubMed]
32. Liu, T.; Qiao, N.; Ning, F.J.; Huang, X.Y.; Luo, L.P. Identification and characterization of plant-derived biomarkers and physicochemical variations in the maturation process of Triadica cochinchinensis honey based on UPLC-QTOF-MS metabolomics analysis. *Food Chem.* **2022**, *13*, 13517. [CrossRef]
33. Barrett, D.P.; Fowler, S.V.; Subbaraj, A.K.; Groenteman, R.; Clavijo-McCormick, A. Metabolomic analysis of host plant biochemistry could improve the effectiveness and safety of classical weed biocontrol. *Biol. Control* **2021**, *160*, 104663. [CrossRef]
34. Hall, R.D.; D'Auria, J.C.; Ferreira, A.C.S.; Gibon, Y.; Kruszka, D.; Mishra, P.; Zedde, R.V.D. High-throughput plant phenotyping: A role for metabolomics? *Trends Plant Sci.* **2022**, *27*, 549–550. [CrossRef]
35. Mithöfer, A.; Boland, W. Plant defense against herbivores: Chemical aspects. *Annu. Rev. Plant Biol.* **2012**, *63*, 431–450. [CrossRef]
36. Pichersky, E.; Lewinsohn, E. Convergent evolution in plant specialized metabolism. *Annu. Rev. Plant Biol.* **2011**, *62*, 549–566. [CrossRef]
37. Chen, F.; Tholl, D.; Bohlmann, J.; Pichersky, E. The family of terpene synthases in plants: A mid-size family of genes for specialized metabolism that is highly diversified throughout the kingdom. *Plant J.* **2011**, *66*, 212–229. [CrossRef] [PubMed]
38. War, A.R.; Buhroo, A.A.; Hussain, B.; Ahmad, T.; Nair, R.M.; Sharma, H.C. Plant defense and insect adaptation with reference to secondary metabolites. In *Co-Evolution of Secondary Metabolites, Reference Series in Phytochemistry*; Springer: Cham, Switzerland, 2020.
39. Uawisetwathana, U.; Graham, S.F.; Kamolsukyunyong, W.; Sukhaket, W.; Klanchui, A.; Toojinda, T.; Vanavichit, A.; Karoonuthaisiri, N.; Rlliott, C.T. Quantitative 1H NMR metabolome profiling of Thai Jasmine rice (*Oryza sativa*) reveals primary metabolic response during brown planthopper infestation. *Metabolomics* **2015**, *11*, 1640–1655. [CrossRef]

40. Peng, L.; Zhao, Y.; Wang, H.Y.; Zhang, J.J.; Song, C.P.; Shuangguan, X.X.; Zhu, L.L.; He, G.C. Comparative metabolomics of the interaction between rice and the brown planthopper. *Metabolomics* **2016**, *12*, 132. [CrossRef]
41. Zhou, S.Q.; Lou, Y.R.; Tzin, V.; Jander, G. Alteration of plant primary metabolism in response to insect herbivory. *Plant Physiol.* **2015**, *169*, 1488–1498. [CrossRef]
42. Coruzzi, G.M.; Last, R.L. Amino Acids. In *Biochemistry and Molecular Biology of Plants*; Buchanan, R.B., Gruissem, W., Jones, R., Eds.; Wiley Press: Rockville, MD, USA, 2000; pp. 358–410.
43. Koyama, Y.; Yao, I.; Akimoto, S.Ä. Aphid galls accumulate high concentrations of amino acids: A support for the nutrition hypothesis for gall formation. *Entomol. Exp. Appl.* **2004**, *113*, 35–44. [CrossRef]
44. Dorschner, K.W.; Ryan, J.D.; Johnson, R.C.; Eikenbary, R.D. Modification of host nitrogen levels by the greenbug (Homoptera: Aphididae): Its role in resistance of winter wheat to aphids. *Environ. Entomol.* **1987**, *16*, 1007–1011. [CrossRef]
45. Sandström, J.; Telang, A.; Moran, N.A. Nutritional enhancement of host plants by aphids–A comparison of three aphid species on grasses. *J. Insect Physiol.* **2000**, *46*, 33–40. [CrossRef]
46. Zeng, L.; Zhou, X.C.; Liao, Y.Y.; Yang, Z.Y. Roles of specialized metabolites in biological function and environmental adaptability of tea plant (*Camellia sinensis*) as a metabolite studying model. *J. Adv. Res.* **2019**, *34*, 159–171. [CrossRef]
47. Li, X.W.; Zhang, J.; Lin, S.B.; Xing, Y.X.; Zhang, X.Z.; Meng, Y.; Chang, Y.L.; Guo, H.W.; Sun, X.L. (+)-Catechin, epicatechin and epigallocatechin gallate areimportant inducible defensive compounds against Ectropis grisescens in tea plants. *Plant Cell Environ.* **2021**, *2*, 496–511.
48. Wan, X.C.; Xia, T. *Secondary Metabolism of Tea Plant*, 1st ed.; Science Press: Beijing, China, 2015.
49. Zhao, X.M.; Chen, S.; Wang, S.S.; Shan, W.N.; Wang, X.X.; Lin, Y.Z.; Su, F.; Yang, Z.B.; Yu, X.M. Defensive Responses of Tea Plants (*Camellia sinensis*) Against Tea Green Leafhopper Attack: A Multi-Omics Study. *Front. Plant Sci.* **2020**, *10*, 1705. [CrossRef] [PubMed]
50. Maeda, H.; Dudareva, N. The Shikimate Pathway and Aromatic Amino Acid Biosynthesis in Plants. *Annu. Rev. Plant Biol.* **2012**, *63*, 73–105. [CrossRef] [PubMed]
51. Genva, M.; Akong, F.O.; Andersson, M.X.; Deleu, M.; Lins, L.; Fauconnier, M.L. New insights into the biosynthesis of esterified oxylipins and their involvement in plant defense and developmental mechanisms. *Phytochem. Rev.* **2019**, *18*, 343–358. [CrossRef]
52. Ling, B.; Wang, G.C.; Ya, J.; Zhang, M.X.; Liang, G.W. Antifeedant activity and active ingredients against *Plutella xylostella* from *Momordica charantia* leaves. *Agr. Sci. China* **2008**, *7*, 1466–1473. [CrossRef]
53. Böttcher, C. *Studies on the Biology of Free and Esterified Phytodienoic Acids*; Ruhr-University Bochum: Bochum, Germany, 2007.
54. Böttcher, C.; Weiler, E.W. Cyclo-oxylipin-galactolipids in plants: Occurrence and dynamics. *Planta* **2007**, *226*, 629–637. [CrossRef]

Disclaimer/Publisher's Note: The statements, opinions and data contained in all publications are solely those of the individual author(s) and contributor(s) and not of MDPI and/or the editor(s). MDPI and/or the editor(s) disclaim responsibility for any injury to people or property resulting from any ideas, methods, instructions or products referred to in the content.